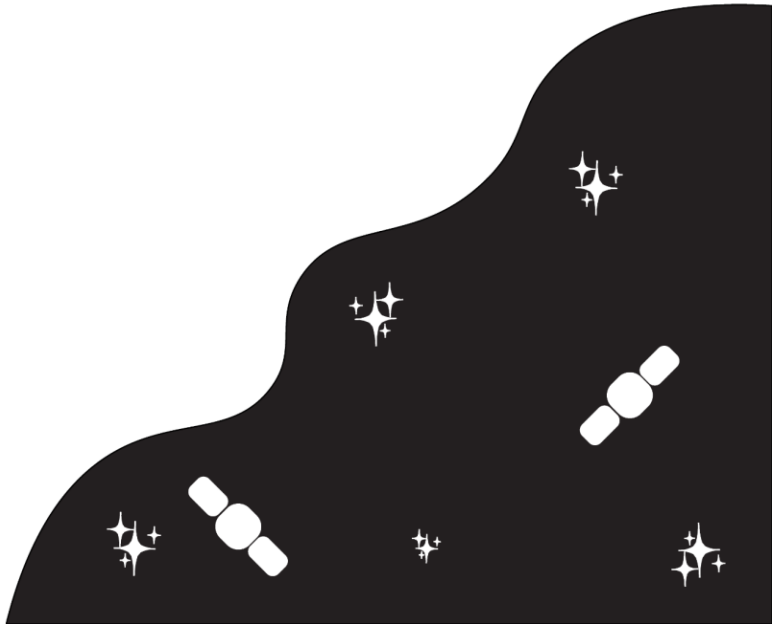
A Review of Early Excess Flux SNe Ia and Their Hosts

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Honours Project A

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Rubin, LSST and Fink

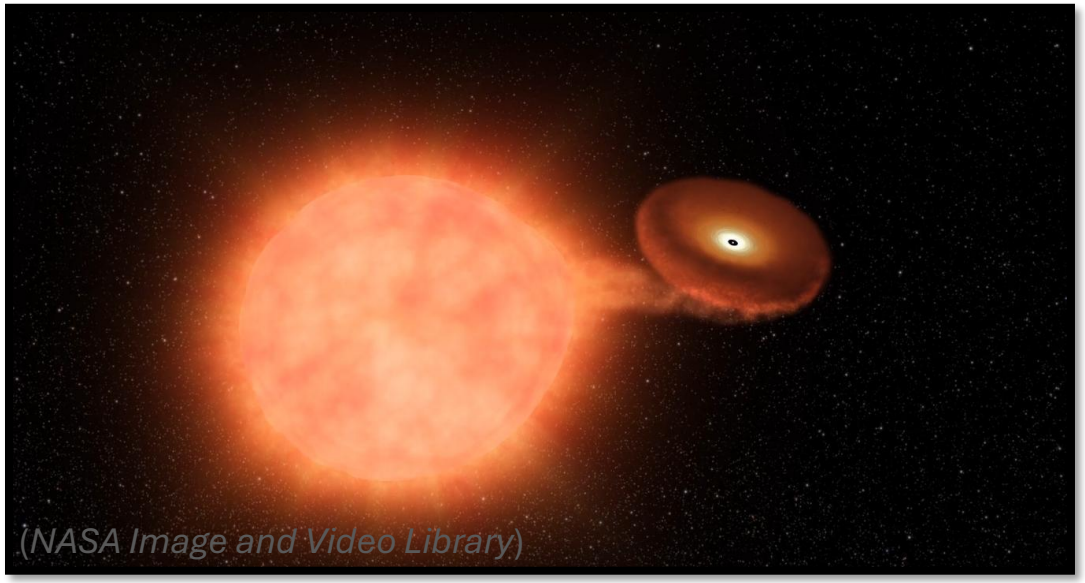
- Rubin telescope
  - 8.36 metre aperture (Bianco et al., 2022)
  - 320.5 – 1084.5 nm (*Vera C. Rubin Observatory Survey Strategy*, n.d.)
- LSST
  - Observe the southern skies.
  - Produces alerts when a change in the sky is detected.
  - Produce 10 million alerts per night, for 10 years. (Möller et al., 2021)
- Fink
  - Filter and classify the alerts.



(Vera C. Rubin Press Release)

What are SN Ia?

- The explosion of a carbon-oxygen white dwarf (Lim et al., 2023)



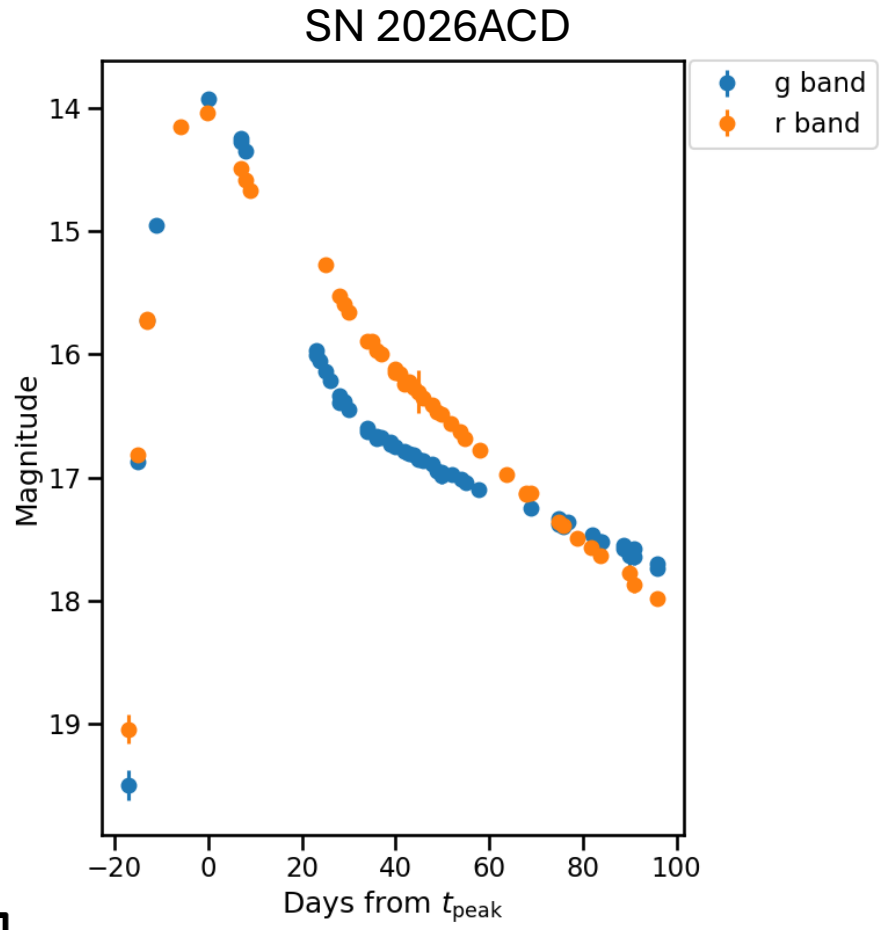
Single Degenerate Scenario



Double Degenerate Scenario

SNe Ia – Why?

- SNe Ia are standardizable candles.
- $\mu = m_B^* - M_B + \alpha x_1 - \beta c$ (Tripp, 1998)
- SALT is used to fit stretch (x_1) and colour (c) (Guy et al., 2005).

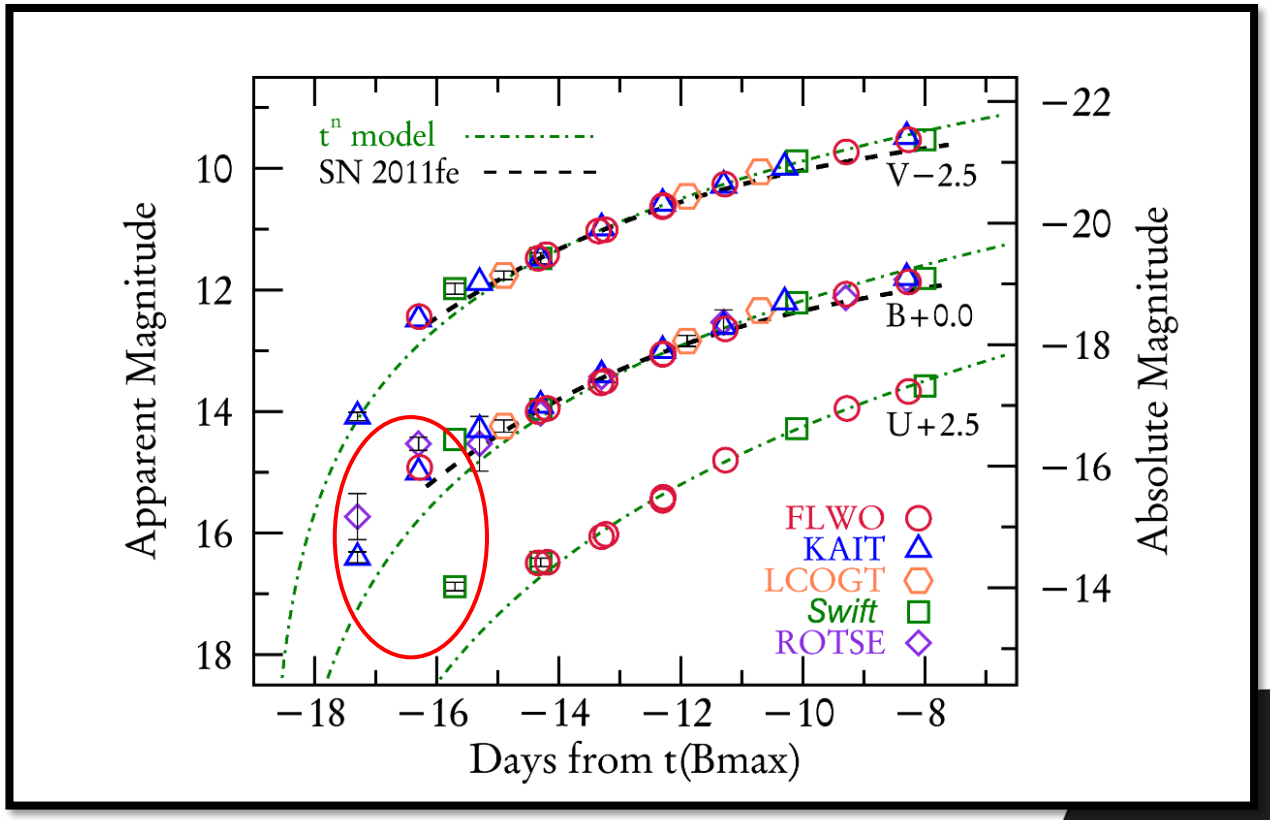


Light curve obtained from the Fink broker.

Early Excess

- In some cases, observation of SNe Ia have a bump in the light curve.

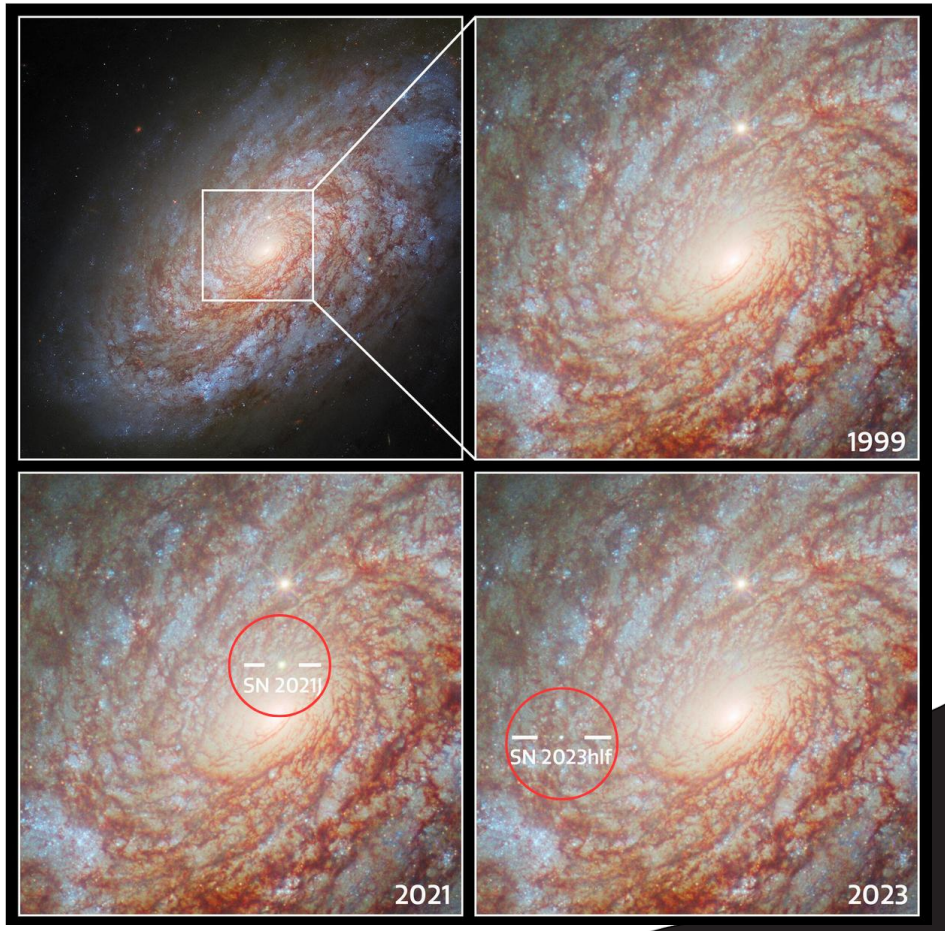
- Reasons For Excess
  - Colour evolution and duration of excess leads to different interpretations (Deckers et al., 2022).
  - Companion interaction (Burke et al., 2022) and (Lim et al., 2023).
  - Clumping of nickel 56 and interactions with CSM (Deckers et al., 2022).



(Marion et al., 2016)

Host Galaxy Environments

- Properties
  - Mass
  - Normalised DLR (separates inner region and outer regions)
  - Star Formation Rate



(*Supernovae in NGC 4414*, n.d.)

Proposed Method

Assemble a catalogue of extragalactic candidates from Rubin using FINK.

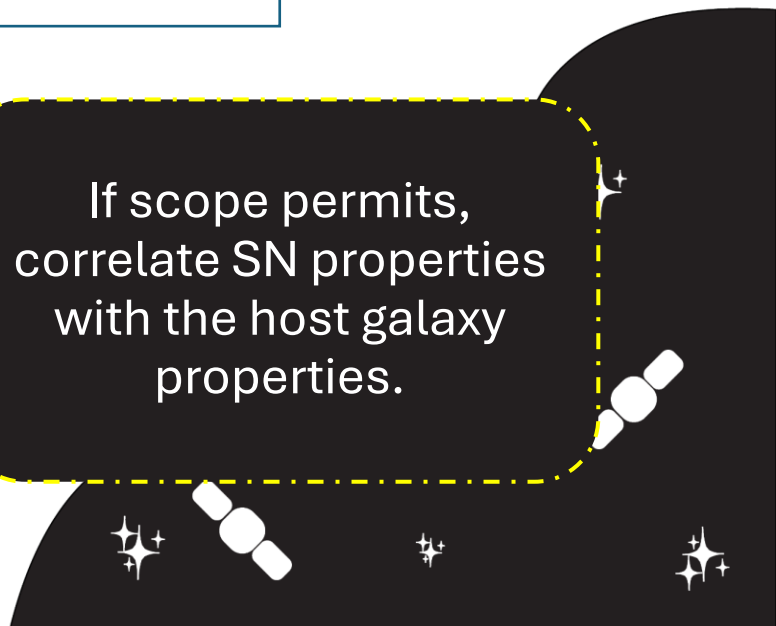
Use ML classifiers to find potential SNe Ia.

Use SALT3 to fit observed light curves (LC) to determine x_1 and c.

Characterise SNe Ia using x_1 and c.

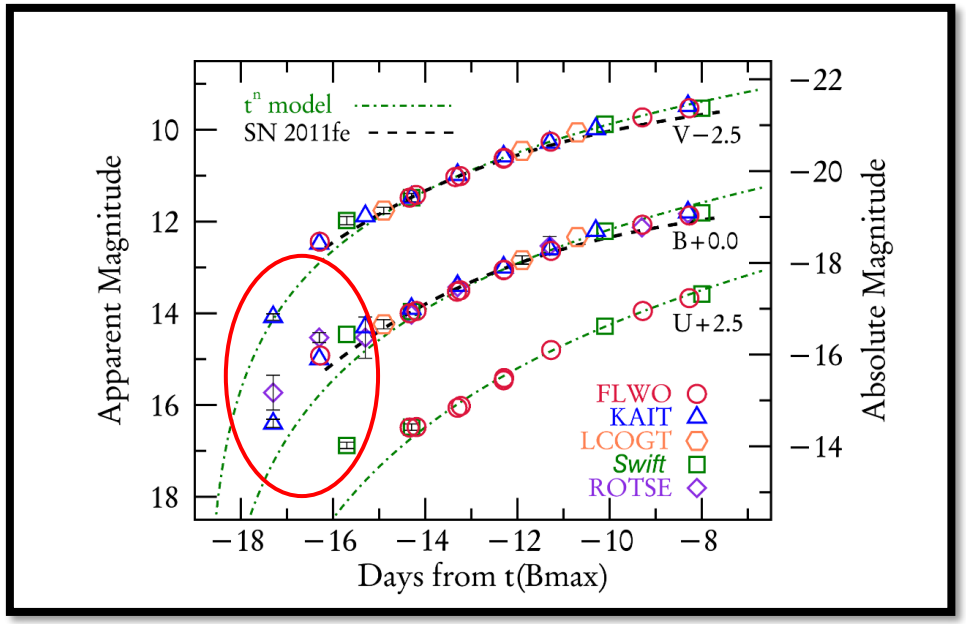
Search for excess flux by finding residuals between fits and observed LC.

If scope permits, correlate SN properties with the host galaxy properties.



Key Takeaways

- Observe, filter and classify transients using Rubin, LSST and Fink.
- Characterise SN using SALT3.
- Search for early excess SN.



SN 2026ACD

